C: > Users > aparn > Downloads > Customer Segmentation.ipynb > import pandas as pd

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[2]

```python
pip install pandas
```

Python

```
Requirement already satisfied: pandas in c:\users\asus\anaconda3\lib\site-packages (2.2.3)
Requirement already satisfied: numpy>=1.26.0 in c:\users\asus\anaconda3\lib\site-packages (from pandas) (2.1.3)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\asus\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\asus\anaconda3\lib\site-packages (from pandas) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\asus\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\asus\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

```python
import pandas as pd
import numpy as np  # Fixed typo: 'ud' to 'np'
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

# 2. Load Dataset
# The error occurs because 'Mall_Customers.csv' file is not found
# Option 1: Provide the correct path to your file
# df = pd.read_csv("path/to/your/Mall_Customers.csv")

# Option 2: Download the dataset if you don't have it
# Uncomment the following lines to download the dataset
from io import StringIO
import requests

# Download Mall_Customers dataset from a reliable source
url = "https://raw.githubusercontent.com/SteffiPeTaffy/machineLearningAZ/master/Machine%20Learning%20A-Z%20Template%20Folder/Part%204%20-%20Clustering/Section%2024%20-%20K-Means%20C
```

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```python
# Download Mall_Customers dataset from a reliable source
url = "https://raw.githubusercontent.com/SteffiPeTaffy/machineLearningAZ/master/Machine%20Learning%20A-Z%20Template%20Folder/Part%204%20-%20Clustering/Section%2
response = requests.get(url)
df = pd.read_csv(StringIO(response.text))

print("First 5 rows:")
print(df.head())
print("\nDataset Info:")
print(df.info())

# 3. Data Preprocessing
# Select important features
x = df[['Annual Income (k$)', 'Spending Score (1-100)']]

# 4. Feature Scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(x)

# 5. Elbow Method to Find K
WCSS = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    WCSS.append(kmeans.inertia_)

plt.figure(figsize=(8,5))
plt.plot(range(1, 11), WCSS, marker='o')
plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()

# 6. Apply K-Means
kmeans = KMeans(n_clusters=5, random_state=42)
clusters = kmeans.fit_predict(X_scaled)
```

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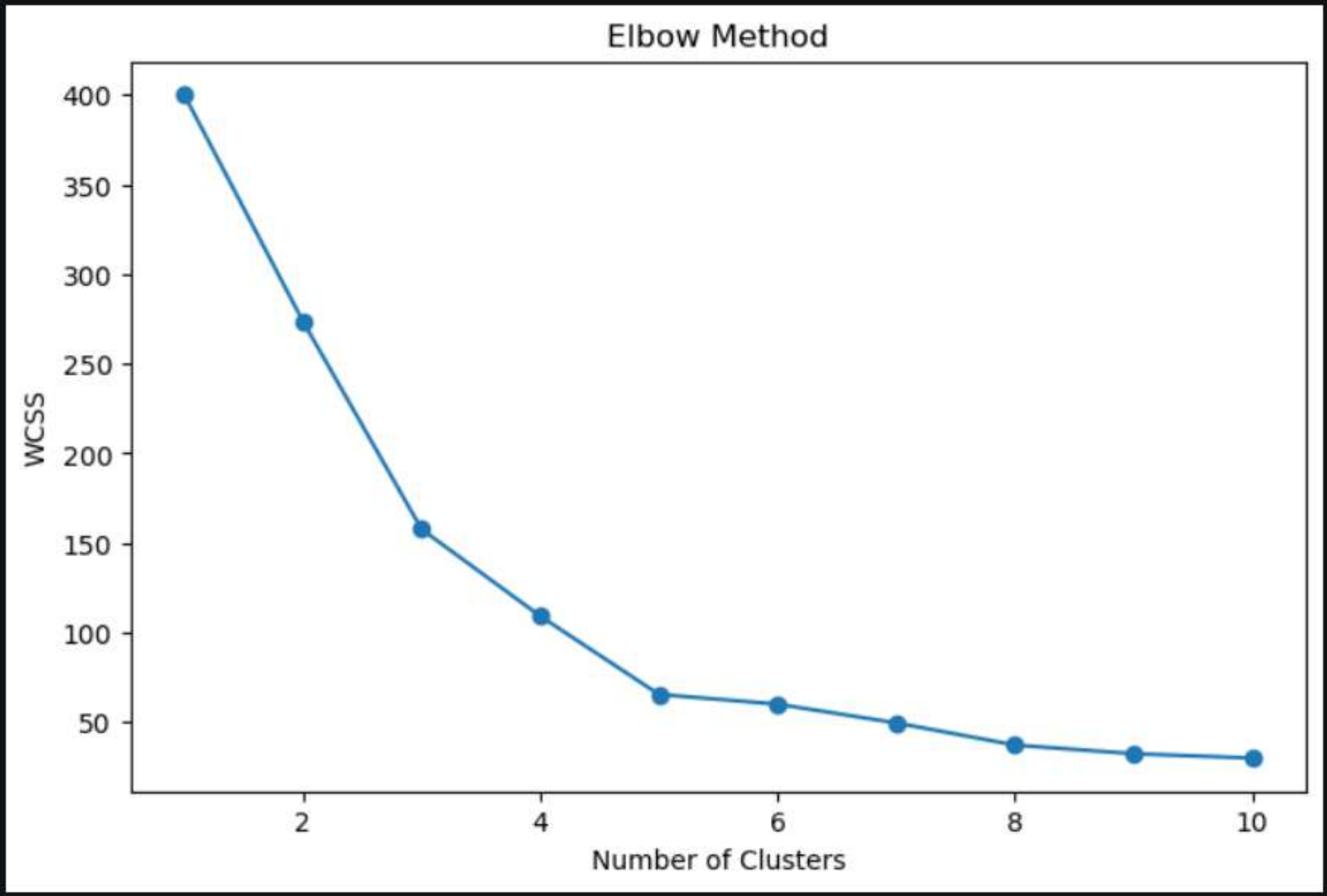
```python
df['Cluster'] = clusters
print("\nClustered Data:")
print(df.head())

# 7. Evaluate Model
score = silhouette_score(X_scaled, clusters)
print("\nSilhouette Score:", score)

# 8. Visualization
plt.figure(figsize=(8,6))
sns.scatterplot(
    x=df['Annual Income (k$)'],
    y=df['Spending Score (1-100)'],
    hue=df['Cluster'],
    palette='viridis'
)
plt.title('Customer Segments')
plt.xlabel('Annual Income')
plt.ylabel('Spending Score')
plt.show()

# 9.Cluster Centers
centers = scaler.inverse_transform(kmeans.cluster_centers_)
print("\nCluster Centers (Original Scale):")
```

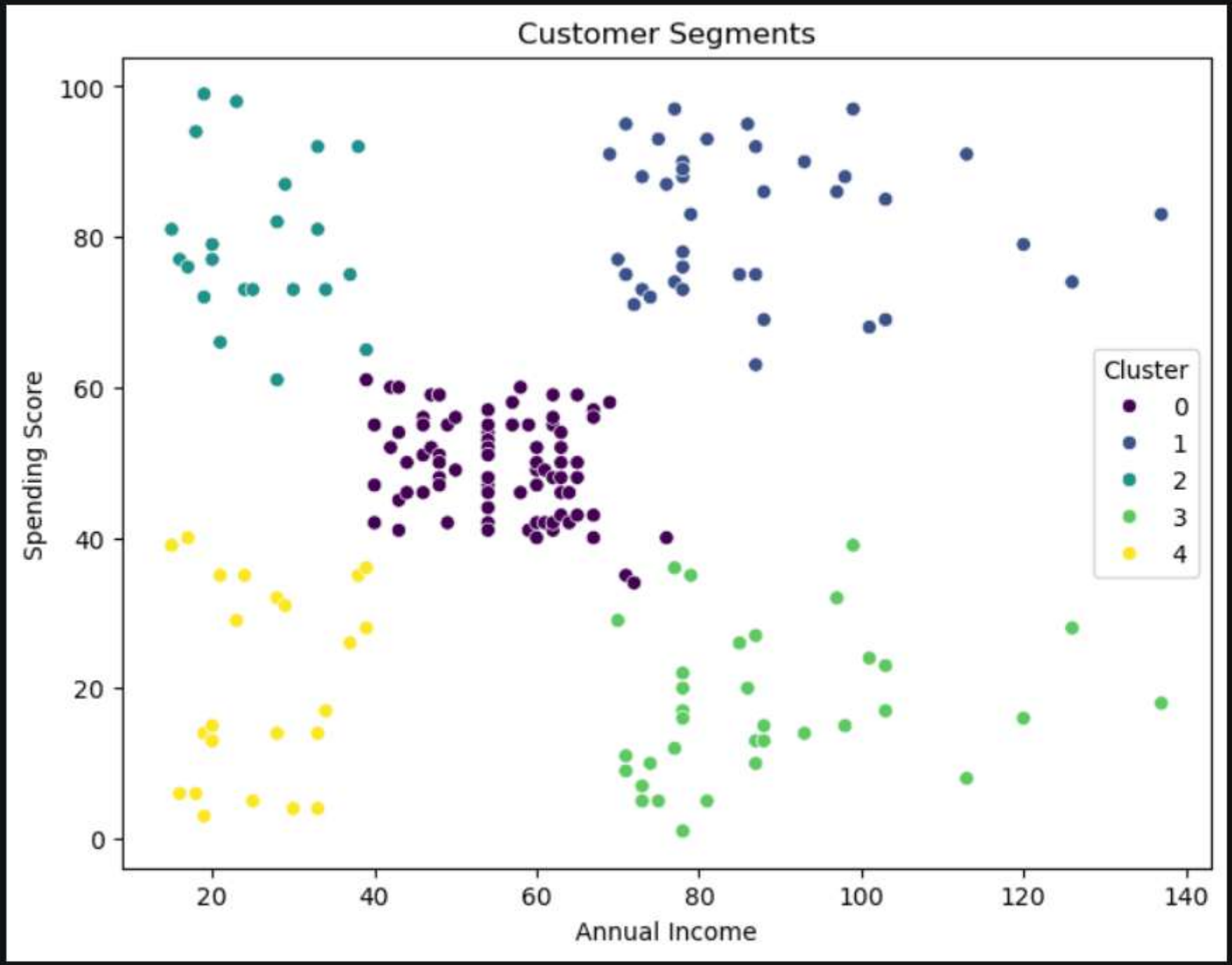
[4] Python

```
First 5 rows:
   CustomerID   Genre  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Genre                   200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
None
C:\Users\Asus\anaconda3\Lib\site-packages\joblib\externals\loky\backend\context.py:136: UserWarning: Could not find the number of physical cores for the following reason:
[WinError 2] The system cannot find the file specified
Returning the number of logical cores instead. You can silence this warning by setting LOKY_MAX_CPU_COUNT to the number of cores you want to use.
  warnings.warn(
  File "C:\Users\Asus\anaconda3\Lib\site-packages\joblib\externals\loky\backend\context.py", line 257, in _count_physical_cores
    cpu_info = subprocess.run(
        "wmic CPU Get NumberOfCores /Format:csv".split(),
        capture_output=True,
        text=True,
    )
  File "C:\Users\Asus\anaconda3\Lib\subprocess.py", line 554, in run
    with Popen(*popenargs, **kwargs) as process:
         ~~~~~^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Asus\anaconda3\Lib\subprocess.py", line 1039, in __init__
    self._execute_child(args, executable, preexec_fn, close_fds,
    ~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
                        pass_fds, cwd, env,
                        ^^^^^^^^^^^^^^^^^^^
```


Elbow Method
400
350
300
250
200
150
100
50
WCSS
2
4
6
8
10
Number of Clusters

```
Clustered Data:
   CustomerID   Genre  Age  Annual Income (k$)  Spending Score (1-100)  \
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40

   Cluster
0        4
1        2
2        4
3        2
4        4

Silhouette Score: 0.5546571631111091
```




```
Cluster Centers (Original Scale):
```